

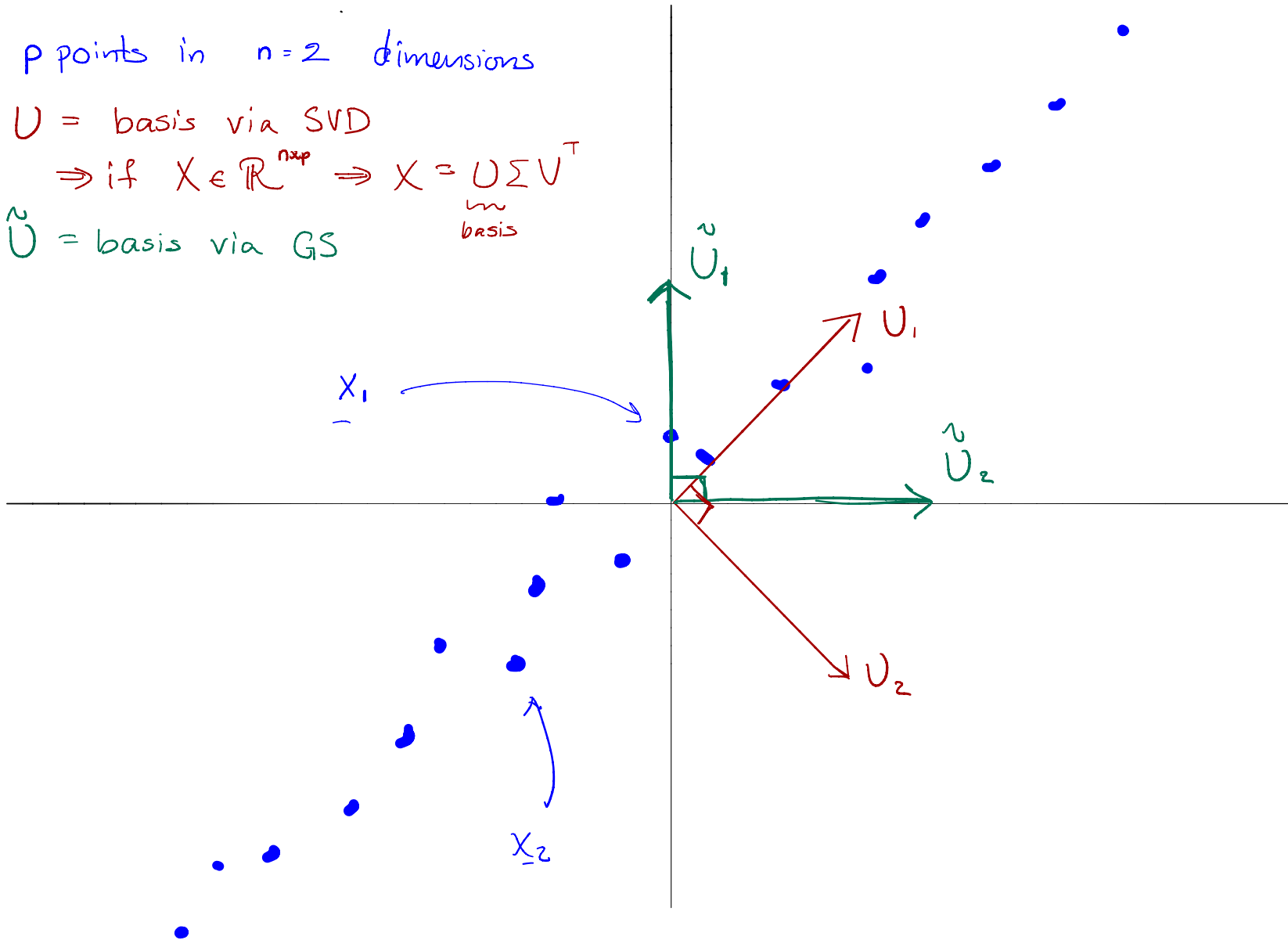
Lecture 7 : Introduction to the Singular Value Decomposition

p points in $n=2$ dimensions

U = basis via SVD

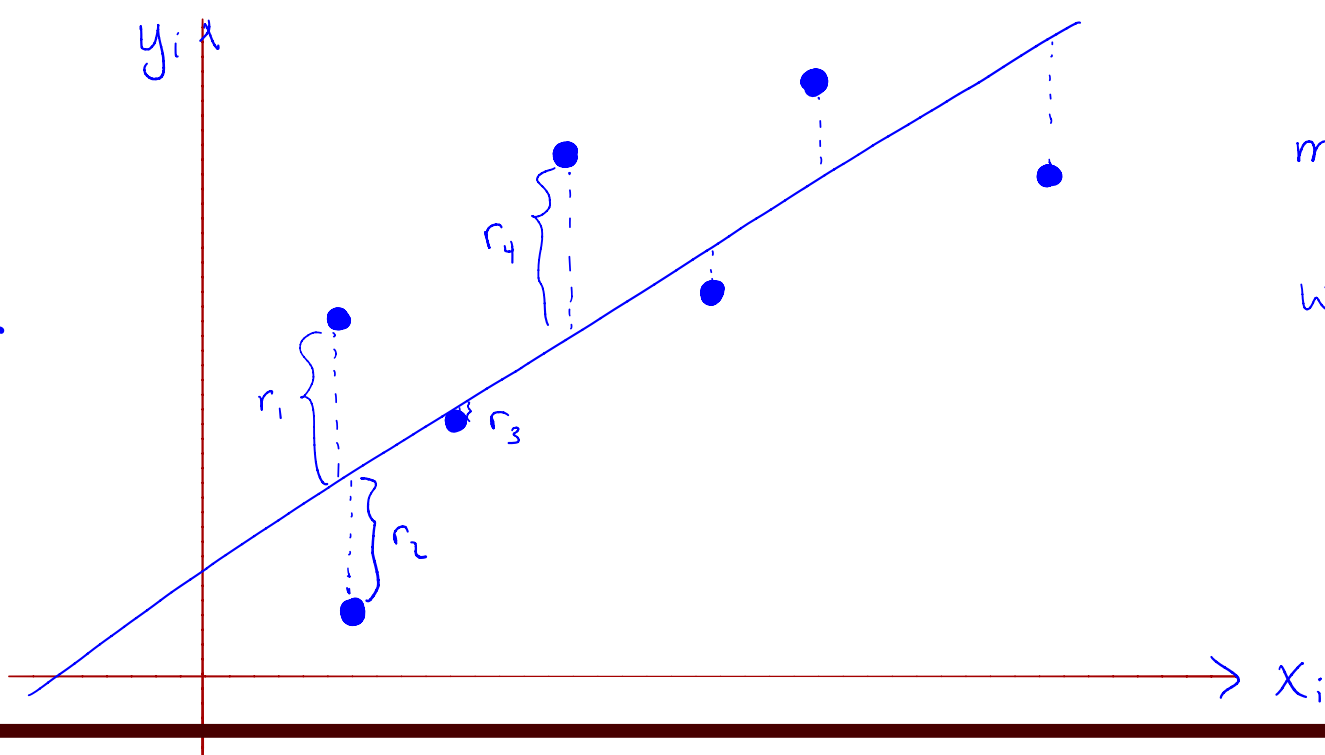
$$\Rightarrow \text{if } X \in \mathbb{R}^{n \times p} \Rightarrow X = \underbrace{U \Sigma V^T}_{\text{basis}}$$

$\hat{U}$ = basis via GS



$\underline{U}_1$ is the 1d subspace that is closest to all the $\underline{X}_i$'s (ie. best 1d subspace fit)

$x_i \in \mathbb{R}$
L.S.



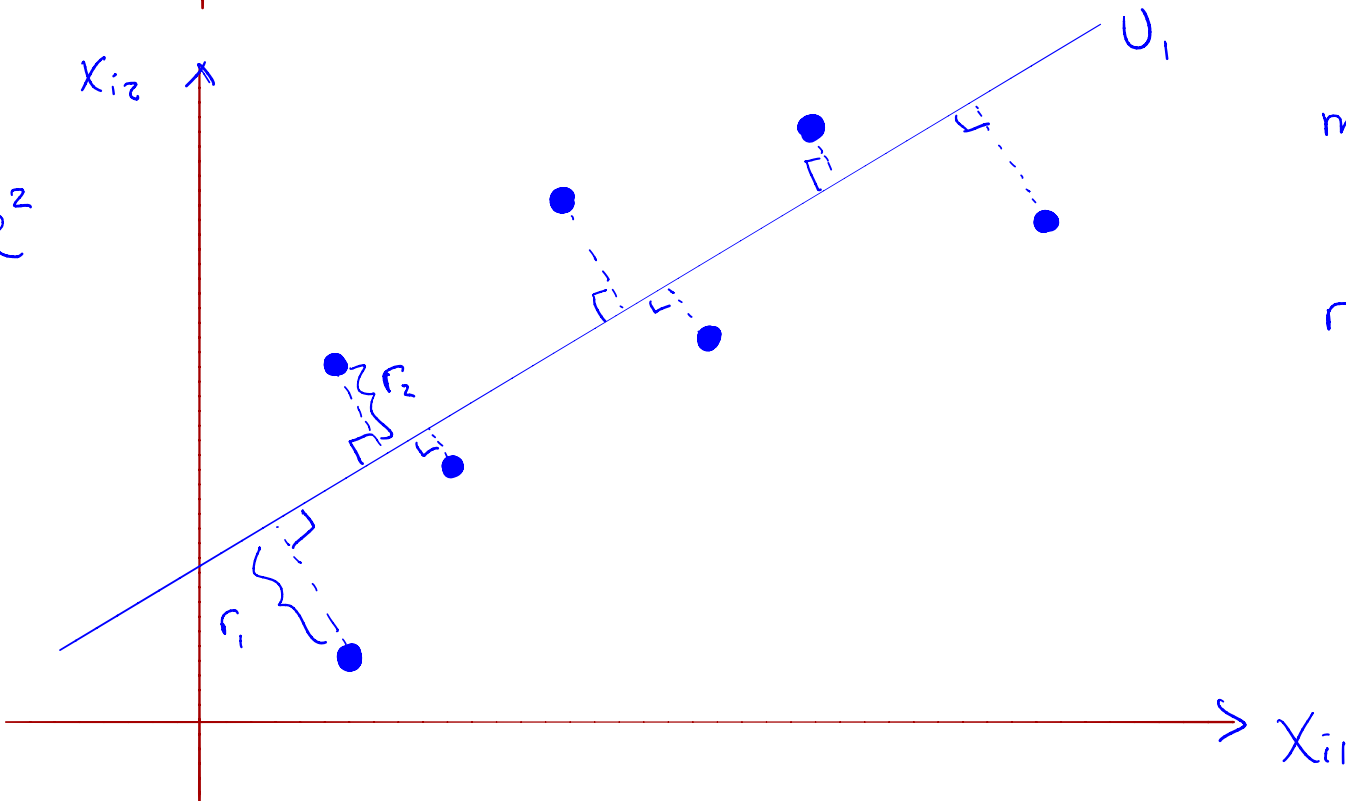
$$\text{minimize } \sum_i r_i^2$$

write

$$y_i \approx ax_i + b$$

$$r_i = y_i - \underbrace{(ax_i + b)}_{\hat{y}_i}$$

$x_i \in \mathbb{R}^2$
SVD

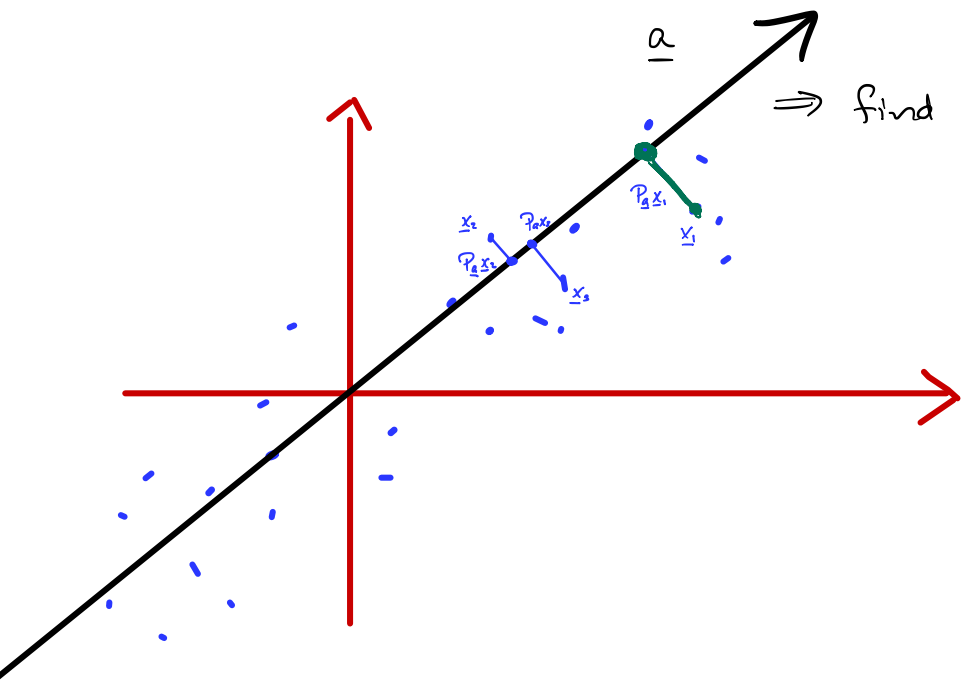


$$\text{minimize } \sum_i \|r_i\|_2^2$$

$$r_i = \underline{x}_i - P_{U_1} \underline{x}_i$$

Introduction to the Singular Value Decomposition (SVD)

Ex. find a 1d-subspace (line through origin) that is closest to a set of points $\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n \in \mathbb{R}^p$



$\Rightarrow$ find $\underline{a}$ to minimize sum of squared distances

Projection Matrices

$$P_A = A(A^T A)^{-1} A^T \Rightarrow P_A = \underline{a}(\underline{a}^T \underline{a})^{-1} \underline{a}^T$$

$$\begin{aligned} P_A^2 &= A(A^T A)^{-1} \cancel{A^T A} (A^T A)^{-1} A^T \\ &= A(A^T A)^{-1} A^T = P_A \end{aligned}$$

given a subspace $\mathcal{S}$ spanned by columns of A ,
the orthogonal complement of $\mathcal{S}$ (aka
orthogonal complement of A) is the set
of all vectors b orthogonal to all columns
of A (i.e. orthogonal to every vector in $\mathcal{S}$)

let $A \in \mathbb{R}^{p \times r}$, $B \in \mathbb{R}^{p \times (p-r)}$ be orthogonal complements
 $\Rightarrow AB = 0$ and any $x \in \mathbb{R}^p$ can be written
as $\underline{x} = \underbrace{A \underline{u}}_{P_A \underline{x}} + \underbrace{B \underline{v}}_{P_B \underline{x}}$ for some $\underline{u} \in \mathbb{R}^r$, $\underline{v} \in \mathbb{R}^{p-r}$

$$\Rightarrow P_A + P_B = I$$

$$\Rightarrow I - P_A = P_A + P_B - P_A = P_B$$

$\Rightarrow I - P_A$ is also a Projection matrix!

distance from $\underline{x}_i$ to line $\underline{a}$:

$$d_i^2 = \|\underline{x}_i - \mathcal{P}_{\underline{a}} \underline{x}_i\|_2^2$$

$$= \|\underline{x}_i - \underline{a}(\underline{a}^T \underline{a})^{-1} \underline{a}^T \underline{x}_i\|_2^2$$

$$= \|\underline{x}_i - \frac{\underline{a} \underline{a}^T}{\underline{a}^T \underline{a}} \underline{x}_i\|_2^2$$

$$= \|(I - \frac{\underline{a} \underline{a}^T}{\underline{a}^T \underline{a}}) \underline{x}_i\|_2^2$$

$$= \underline{x}_i^T (I - \frac{\underline{a} \underline{a}^T}{\underline{a}^T \underline{a}})^T (I - \frac{\underline{a} \underline{a}^T}{\underline{a}^T \underline{a}}) \underline{x}_i$$

$$= \underline{x}_i^T (I - \frac{\underline{a} \underline{a}^T}{\underline{a}^T \underline{a}}) \underline{x}_i$$

$$= \underline{x}_i^T \underline{x}_i - \frac{(\underline{a}^T \underline{x}_i)^2}{\underline{a}^T \underline{a}}$$

$$\text{Want to minimize } \sum_{i=1}^P d_i^2 = \sum_{i=1}^P (\underbrace{\underline{x}_i^T \underline{x}_i}_{\text{constant with respect to } \underline{a}} - \frac{(\underline{a}^T \underline{x}_i)^2}{\underline{a}^T \underline{a}})$$

$$\Rightarrow \arg \min_{\underline{a}} \sum_{i=1}^P d_i^2(\underline{a}) = \arg \max_{\underline{a}} \sum_{i=1}^P \frac{\underline{a}^T \underline{x}_i \underline{x}_i^T \underline{a}}{\underline{a}^T \underline{a}} = \arg \max_{\underline{a}: \underline{a}^T \underline{a} = 1} \sum_{i=1}^P \underline{a}^T \underline{x}_i \underline{x}_i^T \underline{a}$$

$$\text{Let } X = [\underline{x}_1, \underline{x}_2, \dots, \underline{x}_P] \in \mathbb{R}^{n \times P}$$

$$\Rightarrow \underline{u}_1 = \arg \max_{\underline{a}: \underline{a}^T \underline{a} = 1} \sum_i \underline{a}^T \underline{x}_i \underline{x}_i^T \underline{a} = \arg \max_{\underline{a}: \underline{a}^T \underline{a} = 1} \underline{a}^T X X^T \underline{a}$$

value of $\underline{a}$ that achieves maximum is 1st left singular vector of X , denoted $\underline{u}_1$

$$\text{also, } \sigma_1 = \max_{\underline{a}: \underline{a}^T \underline{a} = 1} \|X \underline{a}\|_2 = \|X \underline{u}_1\|_2$$

is called the 1st singular value of X

(also called the "operator norm" of X : $\|X\|_{\text{op}} = \|X\|_2$)

not like 2-norm of vector!

The bigger σ_1 is, the smaller $\sum_i d_i^2$ is

$\Rightarrow$ the better the $\underline{x}_i$'s are aligned with a 1-d subspace

The Singular Value Decomposition

Consider a matrix $X \in \mathbb{R}^{n \times p}$. There always exists matrices U, Σ, V such that

$$X = U \Sigma V^T$$

$U \in \mathbb{R}^{n \times n}$ is orthogonal ($U^T U = U U^T = I$), called left singular vectors
 $V \in \mathbb{R}^{p \times p}$ is orthogonal ($V^T V = V V^T = I$), called right singular vectors
 $\Sigma \in \mathbb{R}^{n \times p}$ is diagonal; diagonal elements called singular values

Σ structure for different dimensions:

$n = p$: $\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}$

$n > p$: $\Sigma = \left[\begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_p \\ \hline & & \end{array} \right] \begin{array}{l} \left. \begin{array}{c} \text{ } \end{array} \right\} p \times p \\ \left. \begin{array}{c} \text{ } \end{array} \right\} (n-p) \times p \end{array}$

$n < p$: $\Sigma = \left[\begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \\ \hline & & \end{array} \right] \begin{array}{l} \underbrace{\hspace{1cm}}_{n \times n} \quad \underbrace{\hspace{1cm}}_{n \times (p-n)} \end{array}$

The columns of U form an orthonormal basis for the columns of X .

The singular values weigh (scale the length) of the corresponding singular vectors.

- The number of non-zero singular vectors is the RANK of X .

The columns of V^T (rows of V) are the basis coefficients (weights on the columns of $U \Sigma$) needed to represent each column of X .

- U gives orthonormal basis for all of $\mathbb{R}^n$.

- 1st r columns of U give basis of best r -dim subspace fit to columns of X

- σ_i 's indicate how important each subspace dimension is to representing/approximating data

- 1st r columns of V give coordinates/locations of each $\underline{x}_i$ within the subspace spanned by $\underline{U}_1, \dots, \underline{U}_r$

- $\underline{U}_1$ = best 1d subspace fit to all data

$$\tilde{\underline{x}}_i^{(1)} = \underline{x}_i - \mathcal{P}_{\underline{U}_1} \underline{x}_i = i^{\text{th}} \text{ residual } \forall i$$

$\underline{U}_2$ = best 1d subspace fit to all $\tilde{\underline{x}}_i^{(1)}$

$\vdots$

$$\tilde{\underline{x}}_i^{(k)} = \underline{x}_i - \mathcal{P}_{[\underline{U}_1, \dots, \underline{U}_k]} \underline{x}_i = \tilde{\underline{x}}_i^{(k-1)} - \mathcal{P}_{\underline{U}_k} \tilde{\underline{x}}_i^{(k-1)} \quad \forall i$$

$\underline{U}_{k+1}$ = best 1d subspace fit to all $\tilde{\underline{x}}_i^{(k)}$

Singular values σ_1, σ_2 , etc. indicate how spread out points are in the subspace.

Recall Gram-Schmidt:

$$\underline{U}_1 = \underline{x}_1 / \|\underline{x}_1\|_2$$

$$\tilde{\underline{x}}_2 = \underline{x}_2 - \mathcal{P}_{\underline{U}_1} \underline{x}_2$$

$$\underline{U}_2 = \tilde{\underline{x}}_2 / \|\tilde{\underline{x}}_2\|_2$$

order of points matters!

basis vectors less interpretable